Mathematical Induction

Proving a statement for every integer, one falling domino at a time.

Greg Ryslik · The Ohio State University

P(k) → P(k+1)

Three Lectures

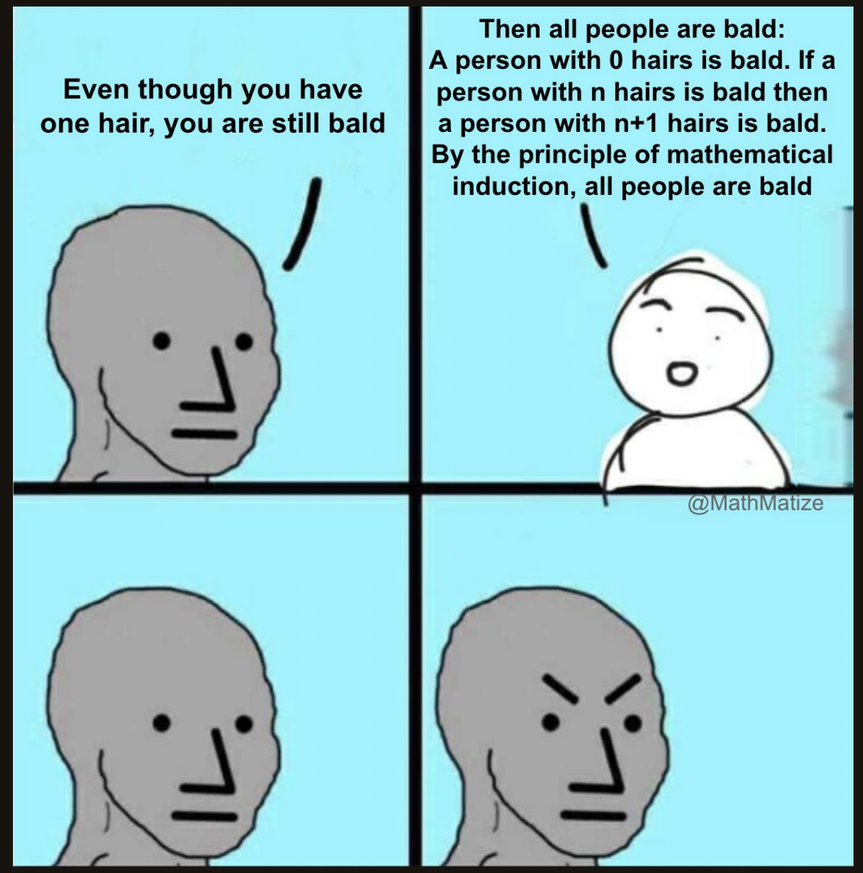
AGENDA

PART I	The principle, a proof template, and sums	LECTURE 08
PART II	Inequalities, and the harmonic numbers	LECTURE 09
PART III	Strong induction, and Fibonacci	LECTURE 10

Every proof in these lectures uses the same four-part template. By the end you should be able to write one without looking.

The Principle

Two premises that knock over infinitely many dominoes.



The Idea: Falling Dominoes

THE IDEA

To knock over an infinite line of dominoes you need exactly **two** things — and then every domino falls.

THINK OF IT AS

Climbing a ladder. You can get onto the first rung, and from any rung you can reach the next one. Then you can reach every rung.

1 Push the first domino over.
BASE CASE

2 Make sure that whenever any domino falls, it knocks over the one after it.
INDUCTIVE STEP

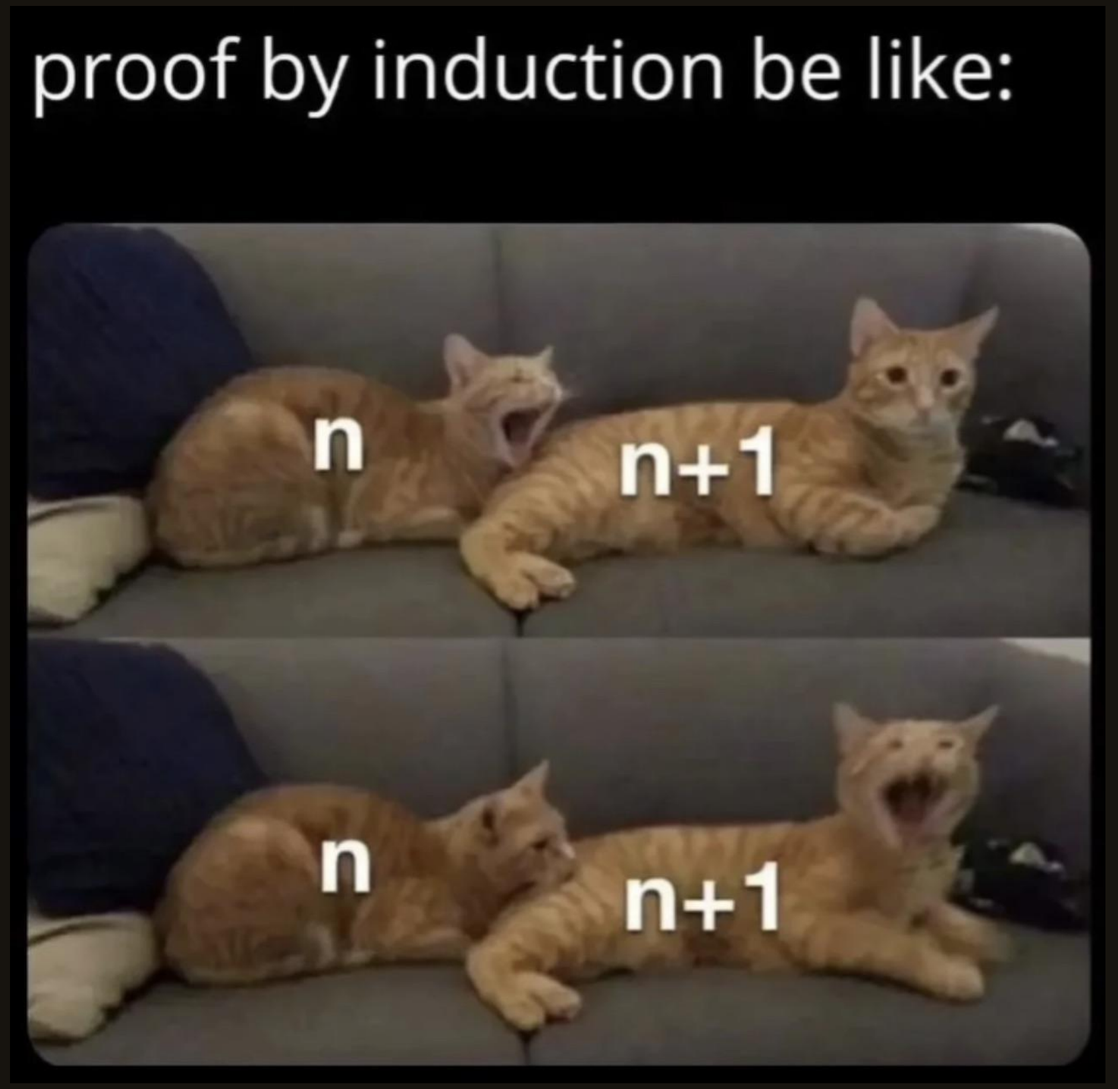
3 Then every domino falls, however far down the line it sits.
CONCLUSION

4 Drop either ingredient and the chain fails. No push: nothing moves. One gap: everything past it stays standing.

BASE CASE + INDUCTIVE STEP

Tip over one domino. Watch them all go.

You push exactly once. Everything after that is the inductive step doing the work.



The Principle of Mathematical Induction

DEFINITION

Let $P(n)$ be a propositional function on the positive integers. The **principle of mathematical induction** is the rule of inference

$$\frac{P(1) \qquad \forall k \geq 1 \ (P(k) \rightarrow P(k+1))}{\therefore \forall n \geq 1 \ P(n)}$$

Two premises, one conclusion — a rule of inference like modus ponens from Lecture 3. $P(k)$ in the second premise is called the **inductive hypothesis**.

The step must hold for **every** $k \geq 1$, including $k = 1$. The base can also start elsewhere: replace 1 by any b and $k \geq 1$ by $k \geq b$.

Why Induction Is Valid

THE INTUITION: UNROLL IT

$P(1)$ base case

$P(1) \to P(2)$ step, $k = 1$

$\therefore P(2)$ modus ponens

$P(2) \to P(3)$ step, $k = 2$

$\therefore P(3)$

⋮

Any particular n is reached after $n - 1$ steps — but no finite chain reaches all of them.

THE PROOF: SMALLEST COUNTEREXAMPLE

Suppose $P(n)$ fails for some n. Every nonempty set of positive integers has a least element (**well-ordering**), so let m be the *smallest* n where P fails.

$m \neq 1$, because $P(1)$ is true. So $m - 1 \geq 1$.

$P(m - 1)$ is true — m was the smallest failure.

The step with $k = m - 1$ gives $P(m)$. **Contradiction.** ■

The unrolling explains why induction feels right. The smallest-counterexample argument is why it is correct.

A Template for Every Induction Proof

01 **State** `P(n)` explicitly, and the starting value `b`.

02 **Base case.** Verify `P(b)` directly — usually one line of arithmetic.

03 **Inductive hypothesis.** Assume `P(k)` for an arbitrary `k ≥ b`. Write it out in full.

04 **Inductive step.** Write the target `P(k+1)` first. Then prove it, and mark the line where the hypothesis is used.

`"By induction, P(n) holds for all n ≥ b." ■`

If the hypothesis never appears in step 4, you did not need induction — or something is wrong.

Example: 1 + 2 + ⋯ + n = n(n+1)/2

n	1 + 2 + ⋯ + n	n(n+1)/2
1	1	1
2	1+2 = 3	3
3	1+2+3 = 6	6
4	⋯ = 10	10
5	⋯ = 15	15

Five for five. That is **evidence** , not proof — there are infinitely many values of n left to check.

A CAUTIONARY TALE

$n^2 + n + 41$ is prime for every n from 0 to 39.

At $n = 40$: $1600 + 40 + 41 = 1681 = 41^2$.

Proof: 1 + 2 + ⋯ + n = n(n+1)/2

EXAMPLE

$P(n)$: $1 + 2 + \cdots + n = n(n+1)/2$, for all $n \geq 1$.

BASE	$P(1)$: $1 = 1 \cdot 2/2$	both sides equal 1 ✓
HYP	$1 + 2 + \cdots + k = k(k+1)/2$	assume for an arbitrary k ≥ 1
GOAL	$1 + \cdots + k + (k+1) = (k+1)(k+2)/2$	P(k+1) — written before we start
STEP	$1 + \cdots + k + (k+1) = k(k+1)/2 + (k+1)$	inductive hypothesis
	$= (k+1)(k/2 + 1)$	factor out k + 1
	$= (k+1)(k+2)/2$	this is P(k+1) ✓

By induction, $P(n)$ holds for every $n \geq 1$. ■ The hypothesis entered exactly once, on the scarlet line.

Practice: Adding Up Odd Numbers

n	$1 + 3 + \cdots + (2n-1)$	sum
1	1	1
2	$1 + 3$	4
3	$1 + 3 + 5$	9
4	$1 + 3 + 5 + 7$	16
5	$1 + 3 + 5 + 7 + 9$	25

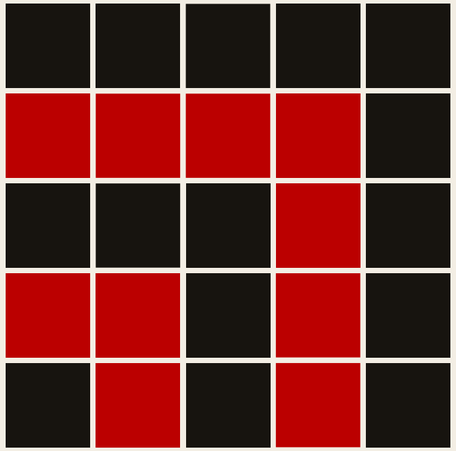
Guess a formula for the sum. Then prove it by induction.

The n-th odd number is $2n - 1$. Use all four parts of the template.

Practice: The Answer

$P(n): 1 + 3 + \cdots + (2n-1) = n^2$, for all $n \geq 1$.

BASE	$1 = 1^2$	✓
HYP	$1 + 3 + \cdots + (2k-1) = k^2$	arbitrary $k \geq 1$
GOAL	$1 + \cdots + (2k-1) + (2k+1) = (k+1)^2$	P(k+1)
STEP	$= k^2 + (2k+1)$	inductive hypothesis
	$= (k+1)^2$	perfect square ✓



Each odd number is an L-shaped layer: $k + k + 1 = 2k + 1$ cells turn a $k \times k$ square into a $(k+1) \times (k+1)$ one.

Example: $1 + 2 + 4 + \cdots + 2^n = 2^{n+1} - 1$

BASE	$n = 0$: $1 = 2^1 - 1$	✓
HYP	$1 + 2 + \cdots + 2^k = 2^{k+1} - 1$	arbitrary $k \geq 0$
GOAL	$1 + \cdots + 2^k + 2^{k+1} = 2^{k+2} - 1$	P(k+1)
STEP	$= (2^{k+1} - 1) + 2^{k+1}$	inductive hypothesis
	$= 2 \cdot 2^{k+1} - 1$	two copies of 2^{k+1}
	$= 2^{k+2} - 1$	✓

SEEN IN BINARY

$1 + 2 + 4 + 8 = 15$

1111_2

$+ \ 1_2$

$10000_2 = 16$

Adding 1 to a string of ones carries all the way. That is why an unsigned k-bit integer tops out at $2^k - 1$.

Sums Worth Knowing

EXAMPLE

sum	closed form	growth
$1 + 2 + \cdots + n$	$n(n+1)/2$	$\Theta(n^2)$
$1^2 + 2^2 + \cdots + n^2$	$n(n+1)(2n+1)/6$	$\Theta(n^3)$
$1^3 + 2^3 + \cdots + n^3$	$n^2(n+1)^2/4$	$\Theta(n^4)$
$1^k + 2^k + \cdots + n^k$	—	$\Theta(n^{k+1})$
$a + ar + \cdots + ar^n$	$a(r^{n+1} - 1)/(r - 1)$	$\Theta(r^n)$ if $r > 1$

Induction **verifies** a formula you already have. It does not tell you where the formula came from.

The geometric row needs $a > 0$ and $r > 1$. For $0 < r < 1$ the sum stays below $a/(1 - r)$, so it is $\Theta(1)$.

Sum of Squares: Check It First

EXAMPLE

n	$1^2 + \cdots + n^2$	$n(n+1)(2n+1)/6$
1	1	$1 \cdot 2 \cdot 3/6 = 1$
2	$1 + 4 = 5$	$2 \cdot 3 \cdot 5/6 = 5$
3	$5 + 9 = 14$	$3 \cdot 4 \cdot 7/6 = 14$
4	$14 + 16 = 30$	$4 \cdot 5 \cdot 9/6 = 30$

$$P(n): 1^2 + 2^2 + \cdots + n^2 = n(n+1)(2n+1)/6$$

Four for four. Now the proof — and this time the algebra in the step takes some care.

Proof: Sum of Squares

EXAMPLE

BASE	$1^2 = (1)(2)(3)/6$	both sides equal 1 ✓
HYP	$1^2 + \cdots + k^2 = k(k+1)(2k+1)/6$	arbitrary $k \geq 1$
GOAL	$1^2 + \cdots + (k+1)^2 = (k+1)(k+2)(2k+3)/6$	P(k+1)
STEP	$1^2 + \cdots + k^2 + (k+1)^2 = k(k+1)(2k+1)/6 + (k+1)^2$	inductive hypothesis
	$= (k+1) \cdot [\, k(2k+1) + 6(k+1) \,] / 6$	factor out k + 1; common denominator
	$= (k+1)(2k^2 + 7k + 6) / 6$	expand the bracket
	$= (k+1)(k+2)(2k+3) / 6$	$(k+2)(2k+3) = 2k^2 + 7k + 6$ ✓

Write the target first. Knowing you need a factor of $(k+2)$ is what tells you how to factor the quadratic.

A Proof That All Horses Are the Same Color

`P(n)`: in any group of `n` horses, all the horses are the same color.

BASE	`n = 1`	one horse is the same color as itself ✓
HYP	`any k horses share a color`	arbitrary k ≥ 1
STEP	`take k + 1 horses: #1, #2, …, #(k+1)`	
	`remove #1 → the rest share a color`	hypothesis on k horses
	`remove #(k+1) → the rest share a color`	hypothesis on k horses
	`the groups overlap → all k + 1 match`	■ ?

Every line looks fine. The conclusion is false. Where is the bug?

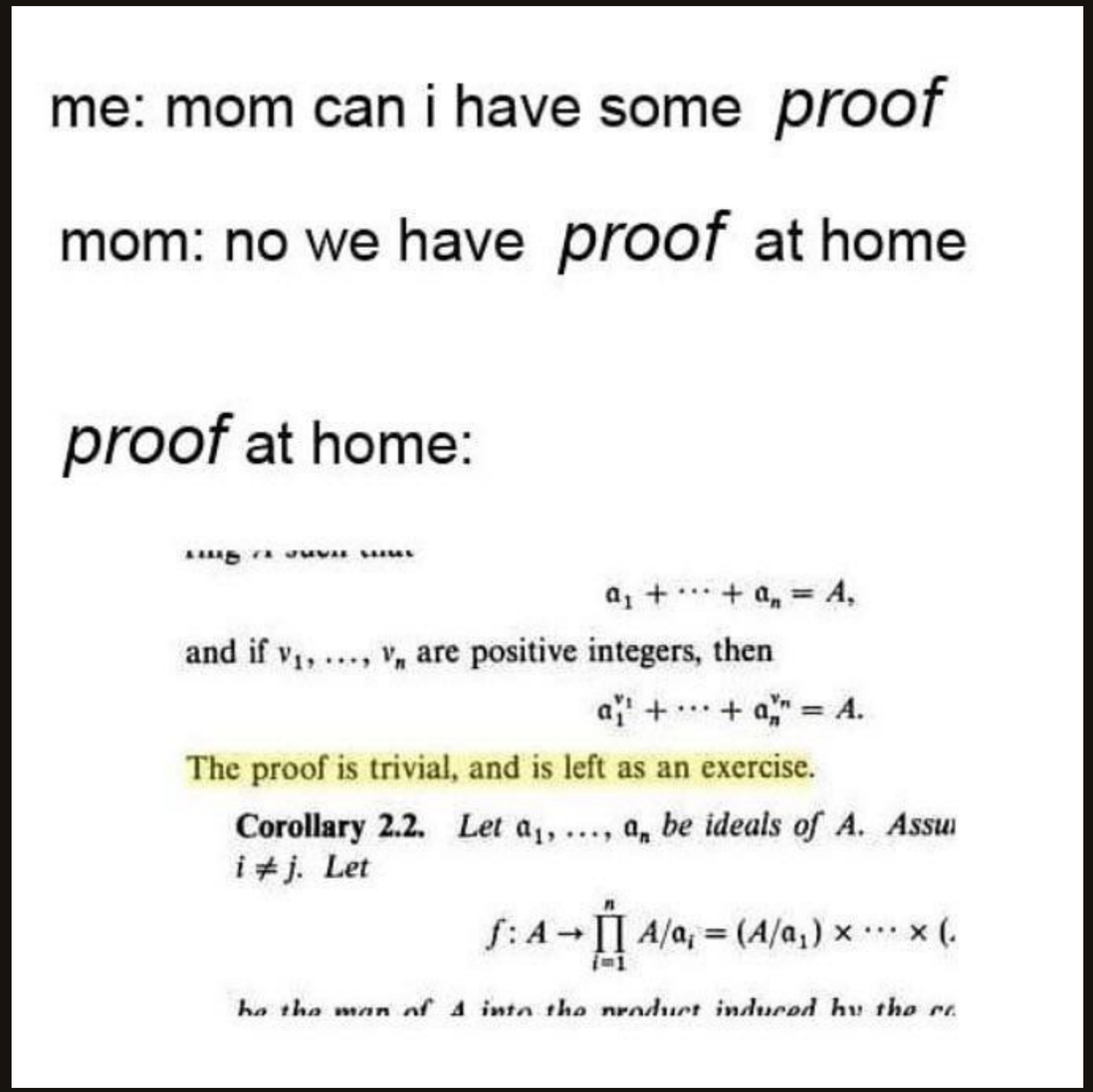
Where It Breaks: k = 1

Run the step with $k = 1$, so $k + 1 = 2$ horses.

REMOVE #1	REMOVE #2	OVERLAP
{ #2 }	{ #1 }	∅

The groups share no horse, so nothing links their colors. The step fails at exactly one value of k — the domino from 1 to 2 is missing, and every domino after it stays standing.

The step does work for every $k \geq 2$. That is what makes the bug hard to see, and why the step must be checked from the **very first** k.



ONE MISSING LINK

Every step checks out. The conclusion is ridiculous.

When a proof reaches something obviously false, the bug is real. Here it hides at k = 1.

Inequalities and Harmonic Numbers

Moving the starting line, and a sum that grows like a logarithm.

Which Grows Faster: 2^n or n^2?

n	2^n	n^2	$2^n > n^2$?
1	2	1	yes
2	4	4	no
3	8	9	no
4	16	16	no
5	32	25	yes
6	64	36	yes
7	128	49	yes

True at $n = 1$, false for $2, 3, 4$, and true from 5 on.

So the claim is $2^n > n^2$ for $n \geq 5$, and the base case is $P(5)$.

The base case goes wherever the pattern actually starts, not automatically at 1.

Proof: $2^n > n^2$ for $n \geq 5$

BASE	$2^5 = 32 > 25 = 5^2$	✓
HYP	$2^k > k^2$	arbitrary $k \geq 5$
GOAL	$2^{k+1} > (k+1)^2$	P(k+1)
STEP	$2^{k+1} = 2 \cdot 2^k$	exponent rule
	$> 2k^2$	hypothesis, times 2
	$= k^2 + k^2$	split
	$\geq k^2 + 2k + 1$	the gap lemma →
	$= (k+1)^2$	✓

THE GAP LEMMA

$k^2 \geq 2k + 1$

$\Leftrightarrow (k - 1)^2 \geq 2$

True once $k - 1 \geq 2$, i.e. $k \geq 3$ — and we have $k \geq 5$.

Inequality steps share a shape: start from the left side of $P(k+1)$, use the hypothesis once, then close the remaining gap with a separate fact.

The Harmonic Numbers

DEFINITION

$$H_n = 1 + 1/2 + 1/3 + \cdots + 1/n$$

Each new term is smaller than the last, so the sum grows — but more and more slowly.

H_n shows up in the average-case analysis of quicksort and hashing, as we will see.

n	H_n	≈
1	1	1.000
2	3/2	1.500
3	11/6	1.833
4	25/12	2.083
5	137/60	2.283
6	49/20	2.450
7	363/140	2.593
8	761/280	2.718

The Idea: Blocks That Each Add ½

Group the terms into blocks that double in length. Every term in a block is at least as big as the block's **last** term.

block	terms	each at least	block sum at least
0	1	—	1
1	1/2	1/2	1/2
2	1/3 + 1/4	1/4	$2 \cdot 1/4 = 1/2$
3	1/5 + ⋯ + 1/8	1/8	$4 \cdot 1/8 = 1/2$
k+1	$1/(2^k+1) + \cdots + 1/2^{k+1}$	$1/2^{k+1}$	$2^k \cdot 1/2^{k+1} = 1/2$

$H_{2^n} \geq 1 + n/2$

Blocks 1 through n each add at least ½ on top of the first term. Induction adds one block per step.

Proof: $H(2^n) \geq 1 + n/2$

$P(n)$: $H_{2^n} \geq 1 + n/2$, for all $n \geq 0$.

BASE	$H_{2^0} = H_1 = 1 \geq 1 + 0/2$	✓
HYP	$H_{2^k} \geq 1 + k/2$	arbitrary $k \geq 0$
GOAL	$H_{2^{k+1}} \geq 1 + (k+1)/2$	P(k+1)
STEP	$H_{2^{k+1}} = H_{2^k} + [\ 1/(2^k+1) + \cdots + 1/2^{k+1}\ ]$	split off the newest block
	$\geq (1 + k/2) + [\ \cdots\ ]$	inductive hypothesis
	$\geq (1 + k/2) + 2^k \cdot 1/2^{k+1}$	2^k terms, each $\geq 1/2^{k+1}$
	$= 1 + (k+1)/2$	✓

Extra Practice

Prove each statement by induction.

(a)

$H_{2^n} \le 1 + n$

for every integer $n \ge 0$. Hint: bound every term in a block *above* by the block's first term.

(b)

$H_1 + H_2 + \cdots + H_n = (n+1)H_n - n$

for every $n \ge 1$. Hint: $H_{k+1} = H_k + 1/(k+1)$.

Sanity check for (b) at $n = 2$: $1 + 3/2 = 5/2$, and $3 \cdot 3/2 - 2 = 5/2$ ✓. Part (a) is used on the next slide.

Theorem: $H_n \in \Theta(\log n)$

We know $1 + m/2 \le H_{2^m} \le 1 + m$. Most n are not powers of two, so **sandwich** them. For $n \ge 2$ let $m = \lfloor \log n \rfloor$, so $2^m \le n < 2^{m+1}$ and $m \ge 1$. H_n increases with n.

LOWER BOUND

$$H_n \ge H_{2^m} \ge 1 + m/2$$
$$> 1 + (\log n - 1)/2$$
$$> \tfrac{1}{2} \log n$$

UPPER BOUND

$$H_n \le H_{2^{m+1}} \le 1 + (m + 1)$$
$$\le \log n + 2$$
$$\le 3 \log n$$

$\frac{1}{2} \log n \le H_n \le 3 \log n$ for all $n \ge 2$: witnesses $c_2 = \frac{1}{2}$, $c_1 = 3$. The last upper step uses $\log n \ge 1$.

Unbounded, but Barely Moving

n	$H_n \approx$
10	2.93
1,000	7.49
1,000,000	14.39
10^9	21.30

H_n grows without bound — but reaching 100 takes about 1.5×10^{43} terms.

WHERE IT SHOWS UP

Quicksort's average number of comparisons is $2(n+1)H_n - 4n$. That is $\Theta(n \log n)$ precisely because $H_n \in \Theta(\log n)$.

DIVERGES, TECHNICALLY

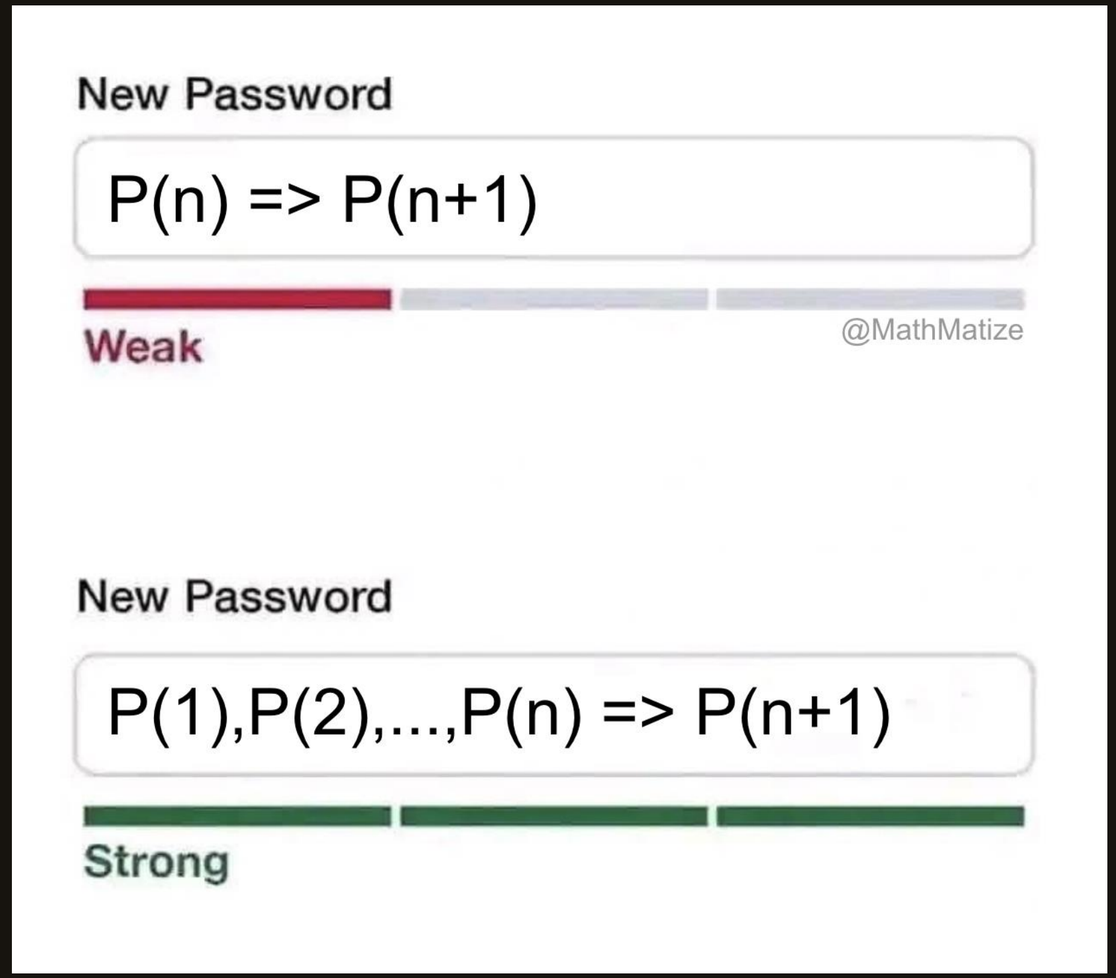
The harmonic series reaches infinity. Please be patient.

After a million terms it has made it to about 14.4.



Strong Induction

When one falling domino isn’t enough to knock over the next.



The Fibonacci Numbers

DEFINITION

$$f_0 = 0 \qquad f_1 = 1 \qquad f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 2$$

n	0	1	2	3	4	5	6	7	8	9	10	11	12
f_n	0	1	1	2	3	5	8	13	21	34	55	89	144

Each term is the sum of the two before it. The definition reaches back **two** steps — and any proof about it has to reach back two steps as well.

When One Domino Isn’t Enough

THE IDEA

Sometimes case `k + 1` needs **several** earlier cases, not just case `k`. So assume all of them.

THINK OF IT AS

A Fibonacci domino is hit by the two before it. Push only the first one and the second has nothing to knock it over — you have to push the first two yourself.

1 Check enough starting cases directly — one for each step back the argument reaches.
BASE CASES

2 Assume the claim for every case from the start up to k.
STRONG HYPOTHESIS

3 Prove case k + 1, using whichever earlier cases you need.
INDUCTIVE STEP

Strong Induction, Formally

DEFINITION

Let $P(n)$ be defined for $n \geq b$. If

$P(b), P(b+1), \ldots, P(b+j)$ are true (the base cases), and

$\forall k \geq b+j \; ([P(b) \wedge \cdots \wedge P(k)] \rightarrow P(k+1))$

then $P(n)$ is true for every $n \geq b$.

The hypothesis now covers **every** earlier case. Use any of them.

Take one base case for each step back the argument reaches. Fibonacci reaches back two, so it needs two.

Strong induction proves nothing ordinary induction cannot: apply ordinary induction to $Q(n) = P(b) \wedge \cdots \wedge P(n)$. It is a convenience, not a new axiom.

One Fact About φ

$\varphi = (1 + \sqrt{5})/2 \approx 1.618$, the golden ratio. Its key property:

$$\varphi^2 = \varphi + 1$$

Multiply both sides by φ^{k-3}: $\varphi^{k-1} = \varphi^{k-2} + \varphi^{k-3}$. Powers of φ obey the Fibonacci recurrence themselves.

$\varphi^2 = ((1 + \sqrt{5})/2)^2$	square it
$= (1 + 2\sqrt{5} + 5)/4$	expand
$= (6 + 2\sqrt{5})/4$	collect
$= (3 + \sqrt{5})/2$	reduce
$= (1 + \sqrt{5})/2 + 1$	split off 1
$= \varphi + 1$	✓

Claim: $f_n > \varphi^{n-2}$ for $n \geq 3$

n	f_n	$\varphi^{n-2} \approx$	$f_n > \varphi^{n-2}$?
2	1	1.000	no — equal
3	2	1.618	yes
4	3	2.618	yes
5	5	4.236	yes
6	8	6.854	yes
7	13	11.090	yes
8	21	17.944	yes

True for every row from $n = 3$ on, and the lead keeps growing.

At $n = 2$ both sides equal 1, so the strict inequality fails. Remember that row.

The two highlighted rows are the base cases we will need.

Proof: $f_n > \varphi^{n-2}$ for $n \geq 3$

BASE	$f_3 = 2 > \varphi^1 \approx 1.618$	n = 3 ✓
BASE	$f_4 = 3 > \varphi^2 \approx 2.618$	n = 4 ✓
HYP	$f_j > \varphi^{j-2}$ for every $3 \leq j \leq k$	strong hypothesis, arbitrary k ≥ 4
GOAL	$f_{k+1} > \varphi^{k-1}$	P(k+1)
STEP	$f_{k+1} = f_k + f_{k-1}$	definition
	$> \varphi^{k-2} + \varphi^{k-3}$	hypothesis on k and k − 1, both ≥ 3
	$= \varphi^{k-3}(\varphi + 1)$	factor
	$= \varphi^{k-3} \cdot \varphi^2$	$\varphi^2 = \varphi + 1$
	$= \varphi^{k-1}$	✓

Why Two Base Cases?

Suppose we checked only $P(3)$ and tried to take the step at $k = 3$, to reach $P(4)$:

$$f_4 = f_3 + f_2 > \varphi^1 + \varphi^0$$ ← needs P(3) and P(2)

$P(2)$: $f_2 > \varphi^0$ means $1 > 1$ — false

The step needs two earlier cases, and at $k = 3$ one of them is outside the claim. So the step starts at $k = 4$, and $P(4)$ must be checked directly.

The rule: one base case for each step back the argument reaches.

What the Bound Tells Us

$\varphi^{n-2} = \varphi^n / \varphi^2$, so $f_n > 0.38 \cdot \varphi^n$.

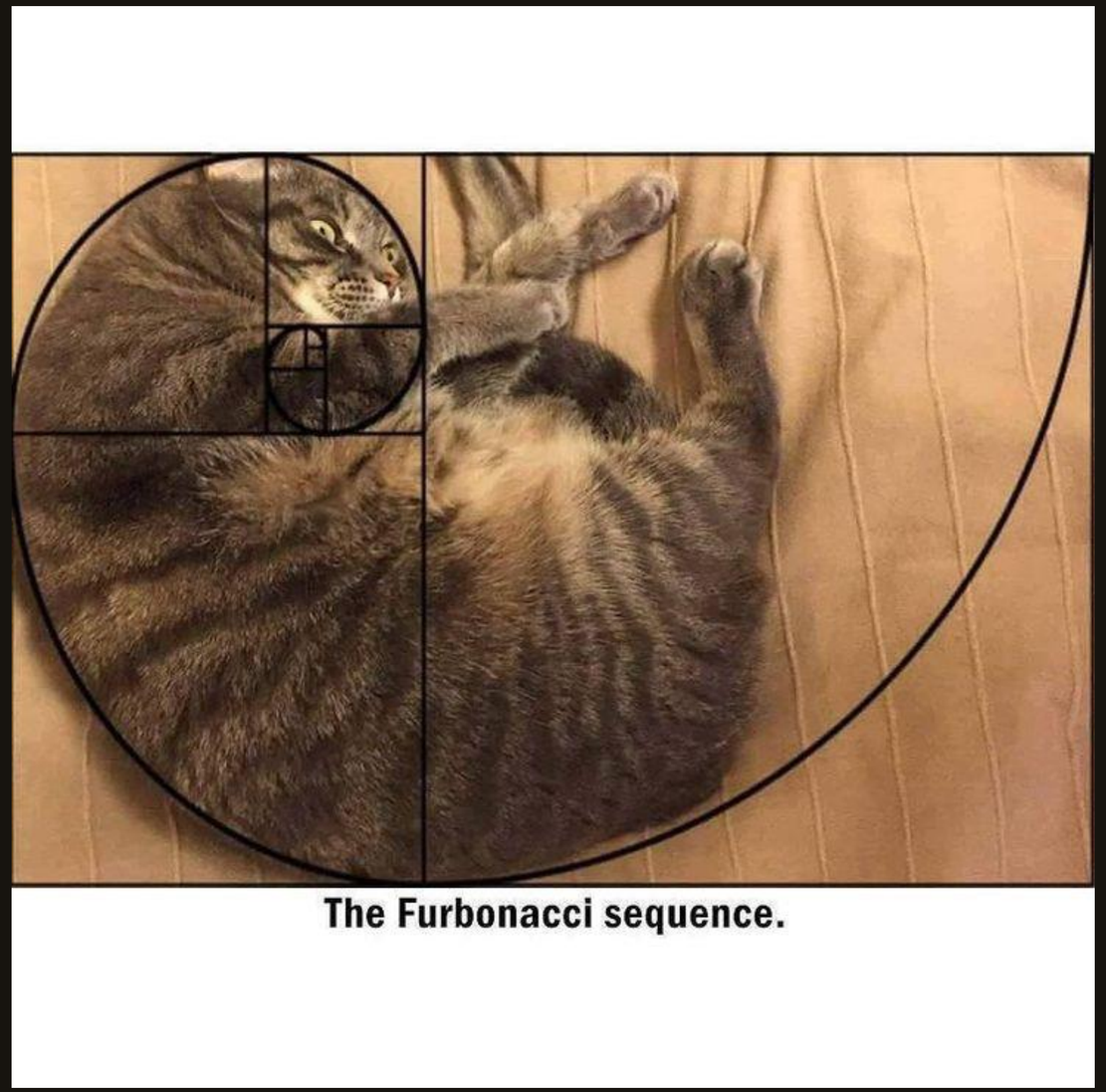
$$f_n \in \Omega(\varphi^n)$$

Exponential growth: $f_{50} = 12{,}586{,}269{,}025$.

THE NAIVE CODE

```
fib(n):
if n < 2: return n
return fib(n-1) + fib(n-2)
```

It makes $2f_{n+1} - 1$ calls. For `fib(50)` that is about **40.7 billion**.



fib(50): 40 BILLION CALLS

Recursive Fibonacci: correct, elegant, and still running.

It recomputes the same values over and over. Remembering answers you already computed fixes it — that’s memoization.

Practice: Postage With 4¢ and 5¢ Stamps

amount	stamps
11	impossible
12	4 + 4 + 4
13	4 + 4 + 5
14	4 + 5 + 5
15	5 + 5 + 5
16	4 + 4 + 4 + 4
17	4 + 4 + 4 + 5

Prove every amount n ≥ 12 can be made from 4¢ and 5¢ stamps.

Compare each highlighted row with the row four above it. How many base cases does that suggest?

Practice: The Answer

$P(n)$: $n = 4a + 5b$ for some integers $a, b \geq 0$, for all $n \geq 12$.

BASE	$12 = 4 \cdot 3$ $13 = 4 \cdot 2 + 5$	n = 12, 13 ✓
BASE	$14 = 4 + 5 \cdot 2$ $15 = 5 \cdot 3$	n = 14, 15 ✓
HYP	$P(j)$ for every $12 \leq j \leq k$	strong hypothesis, arbitrary k ≥ 15
GOAL	$k + 1 = 4a' + 5b'$	P(k+1)
STEP	$k + 1 = (k - 3) + 4$	add one more 4¢ stamp
	$k - 3 = 4a + 5b$	hypothesis: k − 3 ≥ 12 since k ≥ 15
	$k + 1 = 4(a + 1) + 5b$	✓

The step reaches back **four** — from $k + 1$ to $k - 3$ — so there are four base cases.

Common Mistakes

BEFORE YOU GO

01 A step that fails at the smallest k. Check it starting at $k = b$ — the horses fail only at $k = 1$.

02 Assuming the goal. The hypothesis is $P(k)$; $P(k+1)$ is what you prove. Write it down, but never use it.

03 Too few base cases in strong induction. One for each step back the argument reaches.

04 Treating checked cases as a proof. $n^2 + n + 41$ is prime forty times in a row, then isn't.

05 Never using the hypothesis. If it does not appear, the argument is not an induction proof.

Recap

01 Induction needs a base case and a step, and the step must hold for **every** $k \geq b$.

02 Template: state $P(n)$, check the base, write the hypothesis, write the target, and mark where the hypothesis is used.

03 Start where the claim actually becomes true: $2^n > n^2$ starts at 5.

04 $H_n \in \Theta(\log n)$, proved by grouping terms into blocks that double in length.

05 Strong induction assumes every earlier case. $f_n > \varphi^{n-2}$ needs two base cases.

Next: induction returns in loop invariants and in solving recurrences for running time.